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CASE STUDY

In partial fulfilment of the Requirements of the Subject

MM701 Business Analytics Strategy and Practice

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Abstract

This case study looks at how BOB, a leading United Kingdom based online fashion retailer, can apply predictive analytics to boost sales of premium items through targeted email marketing. Though it has 5 million digital consumers by far,, most customers buy within their price range so profit growth has been restricted. The project adopts CRISP-DM framework in order to develop a prediction model identifying the kind of customers who will respond well to up-sales emails. By analyzing customer purchase history, email engagement and behavioral analysis at base, the model sorts out its customers by value on the one hand. It also predicts possible income growth. The results will contribute to BOB making tailored product suggestions, reducing customer discontent, improve mail-marketing results and grow total income. This approach guarantees efficient marketing endeavors with high customer satisfaction and long-term loyalty on the other.

Introduction

One of the popular online fashion brands that run their business in the United Kingdom is **BOB**. The company is a retailer of a broad assortment of clothing, footwear and accessories to millions of customers via its online store.

Bigger digital customer base
BOB has over the years established a huge customer base of almost 5 million registered users. Of them, about 60 percent of clients are sent regular marketing e-mails, where most of them are being offered products that they need or offers that are being made in different seasons or just marketing messages.

Nevertheless, there is a challenge to the company despite its huge network of customers through BOB. Clients tend to purchase items that fall in their normal budget range. To illustrate this point, a client that is used to purchasing £10–£20 T-shirts is not likely to make a purchase of £60 branded hoodie even after being displayed in an email marketing campaign. This behaviour cannot allow BOB to multiply its revenue by selling its products at greater value.

In order to resolve the issue, BOB is interested in applying business analytics to predictive modelling to be aware of the customers who can be more willing to purchase premium or higher-priced products. This is not to give out costly product suggestions to all, but only to those who demonstrate tendencies that might indicate their potential interest of the products in question.

This practice is referred to as **up-selling**.
Up-selling is persuading a customer to purchase something pricier than what he already has in mind or habitually purchases. Up-selling is a feature of most industries:

- Restaurants come with larger meal portions
- Airline companies come with high ticket

- Fashion business with high end products

In the case of BOB, up-selling would significantly boost sales as long as it is carried out appropriately.

This report is aimed at:

- Describing the business problem
- Outlining the analytical goals
- Describing the entire CRISP-DM process
- Identifying the necessary business knowledge
- Explaining how the company will utilize the results
- Explaining how the success of the model is going to be assessed

Everything is explained in very simple English in order to make the ideas simple to grasp.

At the close of this project, BOB would have a system, which utilizes customer purchase history and email engagement behaviour to forecast the respondent customer who would positively act upon the up-selling email messages. This will enable the company to send premium product emails to customers who have the highest probability of making purchases but exclude customers who may respond negatively due to the low-budget products.

Business Objective

The key business goal of this project is:

Increase revenue: A motivational strategy, targeting the identified customers to purchase more expensive goods.

The existing marketing efforts of BOB are irritating a very big customer base with the same kind of email. These mails fail to put into consideration the level of individual customer expenditure, previous purchasing behaviour, or their likes. As a result:

- Customers go on purchasing within their comfort area
- They do not purchase high quality products on a regular basis
- Email marketing does not result in high revenues
- The recommendations cannot be relevant to some customers who just quit opening the emails

It is not only ineffective but also dangerous to send premium product emails to everyone. Consumers that get content they dislike may:

- Feel annoyed
- Lose trust in the brand
- Mark the email as spam or unsubscribe the mail

- Stop thinking about the future marketing activity

When the number of customers unsubscribing or disregarding emails is too high, the functioning of the marketing system of BOB will decline.

Due to these threats, the company does not want to reach the wrong customers. Rather than emailing millions of customers up-selling messages, BOB would like to find smaller but more practical groups of people best suited to purchase premium items.

To do this, BOB requires a predictive model that is able to identify trends on customer behaviour including:

- Spending levels
- Shopping frequencies
- Engagement with emails
- Curiosity towards certain categories
- Reaction to the past campaigns

One such model will enable BOB to create automatically a list of customers that should be promoted to premium products.

Business goal :

We should use the data to make our marketing emails smart enough that only the right people would receive product recommendations of premium items, and BOB would make more profit by spending less on marketing efforts.

This will help in achieving financial objectives of the company as well as long-term satisfaction among customers.

Analytical Goals

The analytics team will be pursuing the following objectives in order to realize the business objective:

Analytical Goal 1: Construct a predictive model

Construct a predictive model, which identifies customers expected to react to up-selling emails.

Several kinds of customer data will be analysed in the predictive model. Certain significant variables are:

- The price bracket of past purchases
- How often the customer shops
- The interval between the last purchase

- Classifications that the customer likes
- Open rate on emails (frequency of email opening)
- Click-through rate (frequency of clicking links on products)
- Customer lifetime value
- Interaction of customer with high value products

Through the analysis of these factors, the model will identify a **high, medium or low propensity** of a customer to give in to the recommendations of premium products.

This will assist BOB in avoiding premium emails to customers that just purchase low-budget products.

Analytical Goal 2: Divide the customers into sections

Customer segmentation is mandatory since customers do not act out there in the same manner. The model will assist BOB to segment the customers into a number of groups including:

- **High value customers:** they make above average purchases
- **Medium-value customers:** they make purchase regularly but not at a high price
- **Low-value customers:** they do not or only buy cheap things
- **Sensitive-customers (price sensitive):** they react to the discounts primarily
- **Premium-interested customers:** they periodically shop or purchase luxury merchandise
- **Loyal customers:** clients that open emails less frequently and who have not made recent purchases

Segmentation assists BOB in sending customised emails to match customer behaviour and preferences.

Analytical Goal 3: Calculate possible growth of revenues

The analytics department will also be used to estimate the amount of the incremental revenue that the model will produce. This includes calculating:

- Anticipated growth in the average order value
- Premium product sales are expected to go up
- Potential income of each of the segments
- Return on Investment (ROI) of utilization of the predictive model

Knowledge of potential revenue growth will assist the management to determine the level of investment in marketing as well as technology.

➤Stages using CRISP-DM framework.

➤ Stage 1: Business Knowledge

Activities:

- Scheduling with marketing departments to get acquainted with objectives
- Determining the key issue: failure in purchasing costly goods by customers
- Defining what "success" means
- Knowledge on customer behaviour and company expectations
- Analytics of old email campaigns and performance
- make sure data is handled legally and ethically.

Expected Results:

- An effective definition of the business issue
- Concurrence of data team and marketing team
- A list of project goals and measure of success
- Awareness of the risk like customer frustration or high levels of unsubscribers

➤ Stage 2: Data Understanding

Activities:

- Accumulation of customer information like purchases, category of products, and order history
- Gathering email interaction information
- Verifying the number of customers who open emails frequently
- Discovering the distribution of the purchase sizes
- Learning the trends like seasonal purchasing behaviour
- Identifying the absence of data or discrepancies
- Production of visualisations which include bar charts, histograms, and line graphs

Expected Results:

- Good knowledge of data that is available
- Awareness of the missing or false data
- Insights such as:
    - Whenever customers open emails more frequently, they are more likely to make purchases
    - Prices of some categories are higher
    - There are customers who do not even premium items
- Early ideas for new features

➤ Stage 3: Data Preparation

Activities:

- Cleaning the dataset
- Removing duplicates
- Handling missing values
- Encoding text categories into numbers
- The development of new options that comprise:
  - Average order value
  - Favourite product category
  - Purchase frequency
  - Email open rate
  - Days since last purchase
  - Price sensitivity score
  - Discount usage pattern
- Normalising data
- Division of data into training and testing data

Expected Results:

- Ready to use clean dataset
- Improved quality of data
- Categories of new variables that can be used to make the model make closely
- Minimized noise and errors of the dataset

➤ Stage 4: Modelling

Activities:

- Applying machine learning algorithms that will help BOB to predict which customers are likely to respond to premium product emails, segment customers based on behavior, and improve prediction accuracy for up-selling campaigns.:
  - Logistic regression
  - Decision trees
  - Random forest
  - Gradient boosting
- Fitting individual models to the data
- Measuring the performance of every model
- Comparison of models based on accuracy, precision and recall
- Choosing the most successful model
- Parameters of the fine-tuning models

Expected Results:

- A strong predictive model
- Good precision and high accuracy
- An up-selling model that defines who the customers are
- A model that is simple to use and comprehend by marketing teams

➤ Stage 5: Evaluation

Required Business Knowledge:

In a business analytics project, it is essential to have specific knowledge at each stage. During the **Business Understanding** stage, one must understand email marketing, pricing strategies, and customer psychology. In the **Data Understanding** stage, knowledge of how customers shop and their seasonal buying patterns is important. The **Data Preparation** stage requires insight into which customer behaviors are significant for analysis. During **Modelling**, understanding the factors that influence customer decision-making is crucial. In the **Evaluation** stage, familiarity with key marketing KPIs helps assess model effectiveness. The **Deployment** stage involves understanding campaign workflows to implement strategies effectively. Finally, during **Monitoring**, interpreting changes in customer interactions ensures that the marketing approach remains relevant and effective.

Activities:

- Mathematical analysis of confusion matrix
- Checking accuracy (the number of customers selected responding)
- Checking recall (number of responsive customers model finds)
- Checking model with unseen new data
- Determining strengths and weaknesses
- Evaluation of model output against those of the business

Expected Results:

- Knowledge of the ability of model to predict customer behaviour
- Improvements in case the model is unsatisfactory
- Indication of the readiness of the model to be deployed

➤ Stage 6: Deployment

Activities:

- How to relate the model to the marketing email system of BOB
- Having an automated customer selection on up-selling campaigns
- Provision of dashboards to the marketing team to monitor outcomes
- Educating employees on the insights of the model

Expected Results:

- Predictive system fully operational
- Precisely automated generation of up-selling email lists
- Quickest and more precise targeting
- The coordination between the marketing and analytics teams will be improved

➤ Stage 7: Monitoring and Review

Activities:

- Post deployment email checking open rates
- Tracking the click rates and conversion rates
- Identifying whether the model degrades with time
- Refining the model with new data on a regular basis
- Customer response analysis based on the long-run

Expected Results:

- Constant improvement of models
- Long-term stability
- Increased income of high quality products
- Improved knowledge of shifting customer behaviour

➤ The Way the Results Will Be Used by the Business

The outcomes of the predictive model will be of great assistance to BOB in a variety of practical directions.

Choosing Customers to Premium Emails:

- The premium product promotions will only be done to the customers who have been categorized as likely responders
- This avoids wasted emails

Recommendations on Products:

- To every customer, it is possible to give suggestions that are related to:
    - Their spending level
    - Their favourite categories
    - Their past purchases
    - Their engagement behaviour
- This enhances client satisfaction

Reducing Customer Irritation:

- Up-selling will not be targeted to customers who do not like high prices or open e-mail very often
- This decreases the unsubscribe rates

Increasing Revenue:
The model will help increase:

- Premium product sales
- Average order value
- Total revenue
- Customer lifetime value

Enhancing Email Marketing Performance:
The KPIs that will be expected to improve marketing will include:

- Open rate
- Click-through rate
- Conversion rate
- Basket value
- Repeat purchase rate

➤ The Evaluation of the Results

The methods that BOB will use to evaluate the model are:

A/B Testing:
Two groups:

- **Group A:** Recipients of model-driven emails
- **Group B:** Recipients of standard emails

The model is effective as demonstrated by better results in Group A only

Revenue Analysis:
BOB will track:

- Growth in average order value
- Revenue from premium items
- Total campaign revenue

Email Engagement Metrics:
Comparing:

- Email open rate
- Click rate
- Unsubscribe rate

Model Performance Metrics:
Using:

- Accuracy - overall correctness.
- Precision - how often predicted buyers are actual buyers.
- Recall - how many actual buyers were identified.
- F1 score - balance of precision and recall.
- Model's Functioning score Model's predictive capability to distinguish between customers likely and unlikely to respond

Customer Retention:

- Following up to know whether the customers remain active after receiving these emails

Long-Term Impact:

- BOB will keep track of whether the customers will keep on buying the more expensive items in the future

➤ Conclusion

This project will give BOB a good evidence-based solution to boost sales by reaching the right customer with high quality products. With predictive modelling and CRISP-DM process, BOB will be able to make smarter marketing decisions that:

- Increase revenue
- Enhance customer satisfaction
- Reduce irrelevant emails
- Enhance customer relationship in the long run
- Increase email response rates

The predictive model will be a valuable resource to the marketing department of BOB and it will help in short term sales campaigns as well as long term business expansion.

➤ References

- Provost, F. & Fawcett, T. (2013). *Data Science for Business*.
- Witten, I., Frank, E. & Hall, M. (2016). *Data Mining: Practical Machine Learning Tools and Techniques*.
- Shmueli, G. et al. (2017). *Data Mining for Business Analytics*.
- Linoff, G.S. and Berry, M.J. (2011) *Data mining techniques: for marketing, sales, and customer relationship management*. 3rd edn. Indianapolis: Wiley.
- Huh, J-H. (2020) 'Predicting customer purchase behaviour using machine learning', *Electronic Commerce Research and Applications*, 39, pp. 1–12.
- Nguyen, T., Zhou, L. and Chen, J. (2022) 'Machine learning models for customer up-selling and cross-selling in e-commerce', *Expert Systems with Applications*, 187, 115889.
- Chapman, P., Clinton, J., Kerber, R., Khabaza, T., Reinartz, T., Shearer, C., & Wirth, R. (2000). *CRISP-DM 1.0: Step-by-step data mining guide*. SPSS Inc. Available at: https://www.crisp-dm.org
- http://khabaza.codimension.net/
- Shearer, C. (2000). *The CRISP-DM model: The new blueprint for data mining*. **Journal of Data Warehousing, 5(4), 13–22**.